

Scan techniques	NMap supports various scan techniques: <code>-sS</code> for a SYN scan; <code>-sP</code> for a ping scan; <code>-sU</code> for a UDP scan; <code>-b</code> for an FTP Bounce scan; and <code>-sI</code> for an idle scan, using a zombie host.
Port specifications	By default, NMap scans the 1000 most common service ports. The <code>--top-ports n</code> switch overrides this default setting with <i>n</i> . The <code>-F</code> switch reduces the most common scanned ports to 100. The following option is very useful if only predetermined ports are required to be scanned: <code>-p U:portnumbers T:portnumbers</code>
Service / version detection	When services are running on non-standard ports, a version detection scan (<code>-sV</code>) on that particular port may provide an excellent option to detect what service it is. To run this scan on all ports, use <code>--all-ports</code> .
Script scan	As mentioned on nmap.org , NSE is NMap's most powerful and flexible feature. Users can write scripts in the Lua programming language for automated scanning. NMap version 5.50 has 177 ready-made NSE scripts in various categories, including discovery, DoS exploits, version-detection and a few more. Some of the intrusive category scripts may crash the target system or use up significant resources on the host.
Operating system detection	The <code>-O</code> option does operating system 'fingerprinting'. NMap version 5.50 has 2982 OS fingerprints and 7319 version-detection signatures.
Timings	Though often neglected, adjusting scan time is very important in effective network scanning. Consider two scenarios: When scanning a large number of hosts, fine-tuning scan timing is essential. For example, scanning a Class B IP address range (up to 65535 addresses) may require significant time. The possible options to use here are: <code>--min-rtt-timeout</code> , <code>--max-rtt-timeout</code> and <code>--host-timeout</code> . To scan devices deployed with IDS/IPS, only a few probes may be allowed in a certain interval; <code>--scan-delay</code> may be used here.
Firewall/IDS evasion and spoofing	<code>-f</code> does fragmentation; <code>-D ip_list</code> sets up decoy hosts. <code>-S ip_addresses</code> does spoofing of source IP address, while <code>--source-port portnumber</code> spoofs source port numbers.
Output	Three basic output options are available: <code>-oN</code> for normal output, <code>-oX</code> for XML output, and <code>-oG</code> for 'greppable' output. The <code>-oA</code> option provides output in all the above formats. One more important option is verbosity. While a scan is running, you may press <code>v</code> to increase verbosity and <code>V</code> to decrease it.
Miscellaneous	The most important is <code>--6</code> , which enables IPv6 scanning.

NMap uses various files to store its default options. Users may edit these files to fine-tune options for individual scanning requirements.

From the exhaustive list above, a few options that come in really handy while scanning networks are shown in Table 2.

Table 2: Interesting NMap options

Option	Description
<code>-sC --smb-users-enum</code> <code>-sC --smb-shares-enum</code>	Really handy for scanning SMB networks, these options respectively return a list of users, and a list of shares detected in the specified host range.
<code>-O</code>	Detects operating systems even of various networking devices.

<code>-A</code>	Runs a standard scan, including OS version detection
<code>-T4</code>	Speeds up the scan; especially useful in quickly scanning a range of IP addresses.
<code>-PN (or -P0)</code>	Ping response is disabled on a few hosts to be scanned. This option assumes the hosts are up/online.

Other tools

The NMap team is also developing some other very interesting tools, some of which are described in Table 3.